

S^1 コンパクト化された時空での リサージェンス構造

森川億人

九大理

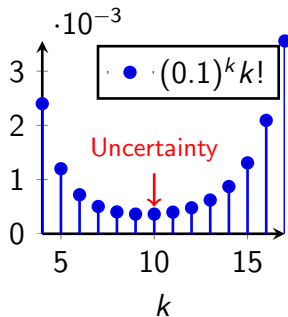
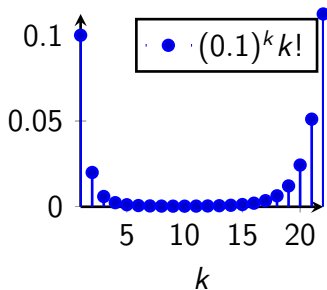
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- O.M. and H. Takaura, PLB **807** (2020) 135570 [arXiv:2003.04759 [hep-th]]
- K. Ishikawa, O.M., K. Shibata and H. Suzuki, PTEP **2020** (2020) 063B02 [arXiv:2001.07302 [hep-th]]
- M. Ashie, O.M., H. Suzuki and H. Takaura, to appear in PTEP [arXiv:2005.07407 [hep-th]]

Factorial growth of perturbation series

- In QM/QFT, perturbative expansions of observables are divergent series
- Typically,

$$F(\lambda) = \sum_{k=0}^{\infty} c_k \lambda^k, \quad c_k \sim k! \text{ at large } k.$$



- Accuracy of perturbative predictions is limited...

Factorial growth of perturbation series

- E.g., ground state energy in QM (Rayleigh–Schrödinger PT):

	Perturbative coefficients
Zeeman effect	$\sim (-1)^k (2k)!$
Stark effect	$\sim (2k)!$
Anharmonic oscillator	
$V(\phi) \sim \phi^3$	$\sim \Gamma(k + 1/2)$
$V(\phi) \sim \phi^4$	$\sim (-1)^k \Gamma(k + 1/2)$
Double well	$\sim k!$
periodic cosine well	$\sim k!$
$\vdots$	$\vdots$

- Due to proliferation of Feynman diagrams (PFD)
 - ▶ # of graphs $\sim k!$

Borel resummation

- The Borel (re)summation is useful for summing divergent asymptotic series.
- For the perturbative series of a quantity $f(g^2)$,

$$f(g^2) \sim \sum_{k=0}^{\infty} f_k \left(\frac{g^2}{16\pi^2} \right)^{k+1},$$

we define the Borel transform by

$$B(u) \equiv \sum_{k=0}^{\infty} \frac{f_k}{k!} u^k.$$

- The Borel sum is given by

$$f(g^2) \equiv \int_0^{\infty} du B(u) e^{-16\pi^2 u/g^2}.$$

Borel resummation

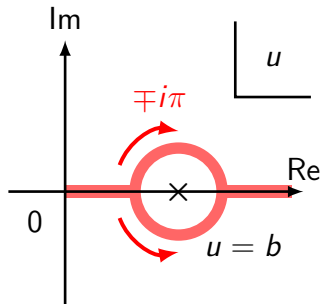
- If $f_k \sim a^k k!$ as $k \rightarrow \infty$,

$$B(u) = \frac{1}{1 - au}.$$

→ Pole singularity at $u = 1/a$ (finite radius of convergence)

- The integral is convergent for $a < 0$ (alternating series).

- If $a > 0$, ill-defined due to the pole (non-Borel summable)
- Avoidance of the pole by contour deformation
- Imaginary ambiguity $\propto$ nonperturbative factor $\sim \pm e^{-16\pi^2/(ag^2)}$



Resurgence theory and semi-classical picture

- $f(g^2)$ is not analytic at $g^2 = 0$, but an asymptotic series.
- Asymptotic nature of perturbative series is related to
 - ① **instability** [Dyson '52, Hurst '52, Thirring '53, ...],
 - ② **nonperturbative effects** such as quantum tunneling [Vainshtein '64, Bender–Wu '73, Lipatov '77, ...]
- Perturbative ambiguity $\propto e^{-\text{const.}/g^2}$

↕ **Cancellation (Resurgence structure)**

Ambiguity associated with **nonperturbative effects**

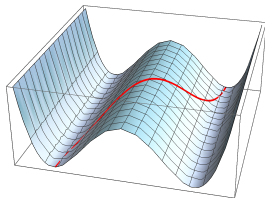
- ▶ PFD vs **Instanton** calculus

[Bogomolny '80, Zinn-Justin '81]

$$\delta_{\text{PFD}} \sim e^{-16\pi^2/g^2}$$



$$\delta_{\text{instanton}} \sim e^{-2S_I} \quad (S_I = 8\pi^2/g^2)$$



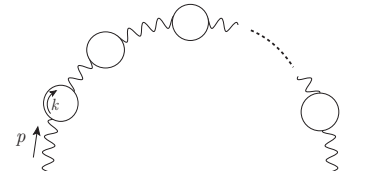
- Reading out nonperturbative effects from PT: **Resurgence theory**

Renormalon and bion

- Another source of $k!$: **renormalon** ['t Hooft 1979]

- ▶ Amplitude of a **single Feynman diagram** $\sim \beta_0^k k!$

E.g., for $SU(N)$ gauge theory,
 $\beta_0 = \frac{11}{3} N$



- Conjecture: renormalon ambiguities disappear thanks to the so-called **bion** [Argyres-Ünsal '12, Dunne-Ünsal '12, ...]
- Bion: a pair of fractional instanton/anti-instanton
 on $\mathbb{R}^{d-1} \times S^1$ with $\mathbb{Z}_N$ -twisted boundary conditions (BC)

	Renormalon	Bion
$\mathbb{R}^4$	$\delta \sim e^{-16\pi^2/(\beta_0 g^2)}$	No
$\mathbb{R}^3 \times S^1$	?	$\delta \sim e^{-2S_B} = e^{-2S_I/N}$

- Renormalon structure under S^1 compactification?

Renormalon analysis on $\mathbb{R}^3 \times S^1$

- Renormalon analysis for 4D $SU(N)$ QCD(adj.) on $\mathbb{R}^3 \times S^1$
($N = 2, 3$ [Anber-Sulejmanpasic '14], $\forall N$ [Ashie-O.M.-Suzuki-Takaura '20])
- (Usually) Renormalon can appear from

$$g^2(\mu^2) \int \frac{d^4 p}{(2\pi)^4} \frac{(p^2)^\alpha}{1 - \Pi(p^2)} \quad \Pi(p^2) \sim g^2(\mu^2) \ln p^2 \quad \text{as } p^2 \rightarrow 0$$

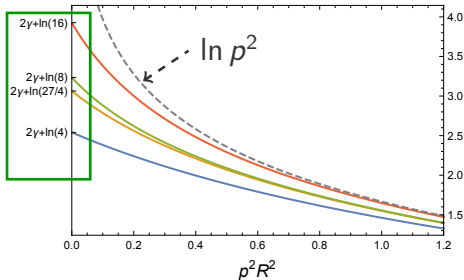
- ▶ Logarithmic factor in vacuum polarization is crucial
- ▶ Coupling expansion: $g^{2(k+1)} \int_p (p^2)^\alpha (\ln p^2)^k \rightarrow g^{2(k+1)} k!$

- In the system on $\mathbb{R}^3 \times S^1$

$$\Pi(p^2 = 0) = \text{const.}$$

- No factorial growth occurs

→ No renormalons



Enhancement of PFD ambiguity and bion

- Inconsistency of conjecture

	Renormalon	Bion
$\mathbb{R}^4$	$\delta \sim e^{-16\pi^2/(\beta_0 g^2)}$	No
$\mathbb{R}^3 \times S^1$	No	$\delta \sim e^{-2S_B} = e^{-2S_I/N}$

- ▶ cf. Renormalon in 2D large- N $\mathbb{C}P^N$ model
⇒ inconsistent with bion [Fujimori-Kamata-Misumi-Nitta-Sakai]

- What cancels bion ambiguities?
- Answer: PFD-type ambiguity (cf. [Anber-Sulejmanpasic])
- Enhancement of PFD upon compactification [O.M.–Takaura '20]

$$\text{PFD: } k! \rightarrow N^k k!$$

$$\delta_{\text{PFD}} \sim e^{-16\pi^2/g^2} \rightarrow \delta'_{\text{PFD}} \sim e^{-16\pi^2/(N g^2)}$$

- Keys (similar to Linde problem ['80] in finite-temperature QFT):
 - ① S^1 compactification $\rightarrow$ IR divergences
 - ② twisted BC $\rightarrow$ twist angles as IR regulators

2-dimensional $\mathbb{C}P^N$ model with twisted BC

- 2D $\mathbb{C}P^{N-1}$ model ($A = 1, 2, \dots, N$)

$$S = \frac{1}{g^2} \int d^2x \left[\partial_\mu \bar{z}^A \partial_\mu z^A - j_\mu j_\mu + f(\bar{z}^A z^A - 1) \right]$$

where $j_\mu = (1/2i) \bar{z}^A \overleftrightarrow{\partial}_\mu z^A$, f is an auxiliary field (Lagrange multiplier for $\bar{z}^A z^A = 1$)

- $\mathbb{Z}_N$ -twisted BC:

$$z^A(x_1, x_2 + 2\pi R) = e^{2\pi i m_A R} z^A(x_1, x_2),$$

where

$$m_A R = \begin{cases} A/N & \text{for } A = 1, 2, \dots, N-1 \\ 0 & \text{for } A = N. \end{cases}$$

- ▶ Periodic BC for f
- ▶ Kaluza–Klein momentum along S^1 : $p_2 = n/R$ ($n \in \mathbb{Z}$)

IR structure of PT: $\mathbb{C}P^N$ model

- Propagator ($f_0 = \langle f \rangle$)

$$\langle z^A(x) \bar{z}^B(y) \rangle = g^2 \delta^{AB} \int \frac{dp_1}{2\pi} \frac{1}{2\pi R} \sum_{p_2=n/R} \frac{e^{i(p_\mu + m_A \delta_{\mu 2})(x-y)_\mu}}{p_1^2 + (p_2 + m_A)^2 + f_0}$$

Significant contribution from $p_2 = 0$, $A = 1$ (mass $(NR)^{-2} + f_0$)

- # of vacuum bubble diagrams ($j_\mu \rightarrow \bar{z}z$)

$$\frac{1}{(2k)!} \left(\frac{\delta}{\delta \bar{z}} \frac{\delta}{\delta z} \right)^{2k} \frac{1}{k!} [(\bar{z}z)^2]^k \sim 4^k \Gamma(k + 1/2) \Rightarrow \text{PFD}$$

- Amplitude of a Feynman diagram (F^{2k} : $2k$ th-order polynomial)

$$\xrightarrow{p_2=0, A=1} \frac{V_2(g^2)^k}{(2\pi R)^{k+1}} \int \left(\prod_{i=1}^{k+1} \frac{dp_{i,1}}{2\pi} \right) \frac{F^{2k}(p_{i,1}, m_A)}{\prod_{i=1}^{2k} [q_{i,1}^2 + m_A^2 + f_0]}$$

IR divergence in massless limit $m_A^2 + f_0 \rightarrow 0$ ($\int d^2p \rightarrow \int dp$)

Enhancement of PFD ambiguity: $\mathbb{C}P^N$ model

- $m_A^2 + f_0 = 1/(NR)^2 + f_0$ works as an IR regulator

- ▶ $m_A^2 \gg f_0$

$$\sim \frac{V_2}{R^2} \frac{1}{(m_A R)^{k-1}} \left(\frac{g^2}{4\pi} \right)^k \xrightarrow{A=1} \frac{V_2}{R^2} \frac{1}{N} \left(\frac{Ng^2}{4\pi} \right)^k \quad \text{Enhancement!}$$

- ▶ $m_A^2 \ll f_0$

$$\sim \frac{V_2}{R^2} \sum_{\alpha \geq 0} \frac{(m_A R)^\alpha}{(f_0 R^2)^{(k+\alpha-1)/2}} \left(\frac{g^2}{4\pi} \right)^k \quad \text{Enhancement}$$

- Dependence on $NR\Lambda$ (Λ : dynamical scale)

- ▶ $NR\Lambda \ll 1$ (Bion calculus is valid)

$$\sqrt{f_0} R \sim \frac{g^2}{4\pi} \Rightarrow [m_A^2 = \mathcal{O}(g^0)] \gg [f_0 = \mathcal{O}(g^4)] \quad \text{Enhancement!}$$

- ▶ $NR\Lambda \gg 1$ (Large N)

$$f_0 \sim \Lambda^2 \Rightarrow \frac{m_A^2}{f_0} = \frac{1}{(NR\Lambda)^2} \ll 1 \quad \text{Enhancement}$$

Summary

- Resurgence structure on $\mathbb{R}^{d-1} \times S^1$

	PT	Semi-classical object
$\mathbb{R}^4$	$\delta_{\text{PFD}} \sim e^{-16\pi^2/g^2}$	$\delta_{\text{instanton}} \sim e^{-2S_I}$
$\mathbb{R}^3 \times S^1$	$\delta'_{\text{PFD}} \sim e^{-16\pi^2/(Ng^2)}$	$\delta_{\text{bion}} \sim e^{-2S_I/N}$
$\mathbb{R}^3 \times S^1$	$\delta_{\text{renormalon}} = 0$	No

- Enhancement phenomenon is consistent with bion
- $NRA \gg 1$ [Ishikawa-O.M.-Shibata-Suzuki '20]
vs $NRA \ll 1$ [Fujimori-Kamata-Misumi-Nitta-Sakai '18]

$$E_{\text{large } N} \sim N^{-1}(\Lambda R)^{-2}\delta\epsilon^2 \quad \Rightarrow f_0 = \Lambda^2 \text{ in denominator}$$

$$\updownarrow$$

$$\downarrow (m_A^2 \text{ vs } f_0)$$

$$\text{Im } E_{\text{bion}} \sim \pm N(\Lambda R)^2\delta\epsilon^2 \quad \Leftarrow \text{enhancement of } N$$

- Emergence of renormalon under decompactification
[Ashie-O.M.-Suzuki-Takaura '20]

- Helpful in giving a unified understanding on resurgence